Introducing .NET MAUI

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David Ortinau

An Introduction via Personal Timeline

‘96-99	HTML/JS, media streaming, ASP, SQL
’99-02	Flash, ASP, SQL
’02-07	.NET, SQL, Flash, Flex
’07-10	Rails, Django, PHP, Flash, Flex, Silverlight, iOS, Android
’11-13	MonoTouch, MonoDroid, iOS, Android, Flash, Flex
‘14-17	Xamarin, Flash, Flex
’17-21	Joined Microsoft as Program Manager for Xamarin.Forms
2021-	.NET MAUI



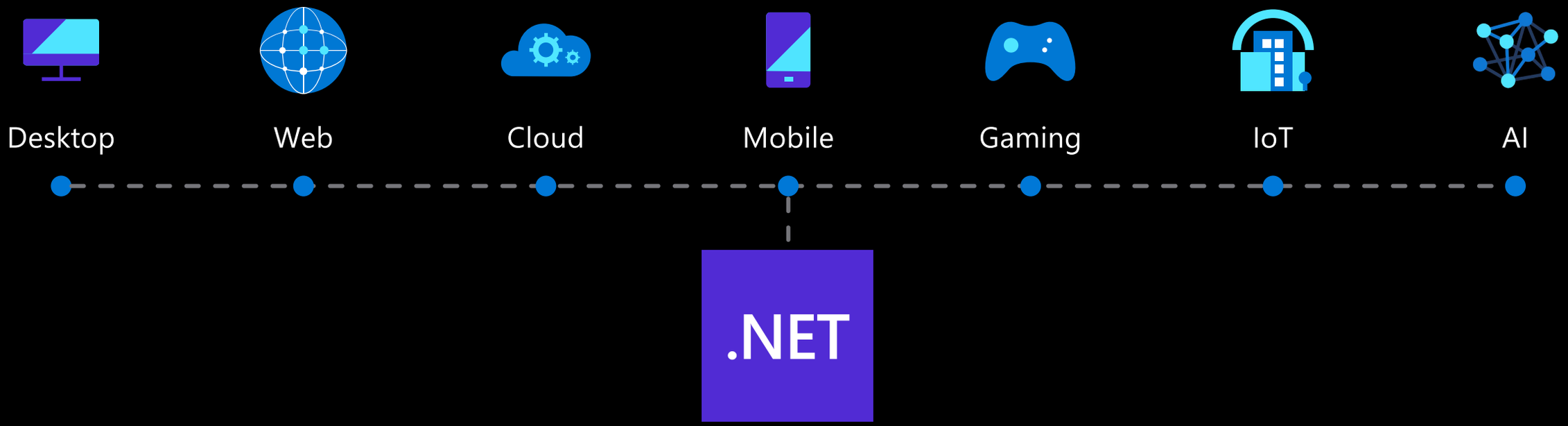
Questions/comments/rants?

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.NET

Your platform for building anything



.NET – A unified development platform



CLOUD

Azure



WEB

ASP.NET

Blazor



DESKTOP

.NET MAUI

WPF

WinForms



MOBILE

.NET MAUI

Xamarin



GAMING

Unity



IoT

ARM32

ARM64



AI

ML.NET

.NET for Apache Spark

.NET 6

COMMON BASE LIBARIES/APIs

INFRASTRUCTURE

RUNTIME COMPONENTS | COMPILERS | LANGUAGES

ECOSYSTEM



NuGet



GitHub



Components, tools, library vendors

TOOLS



Visual Studio



Visual Studio for Mac



Visual Studio Code



CLI

.NET ecosystem momentum

5.3M

Monthly active .NET developers

In Visual Studio family

#1

Most Loved Framework (2019, 2020, 2021)

.NET Core

Top 30

Highest velocity OSS project

github.com/dotnet

Top 5

Language on GitHub

C#

>10X

Faster than Node.js

ASP.NET Core

40%

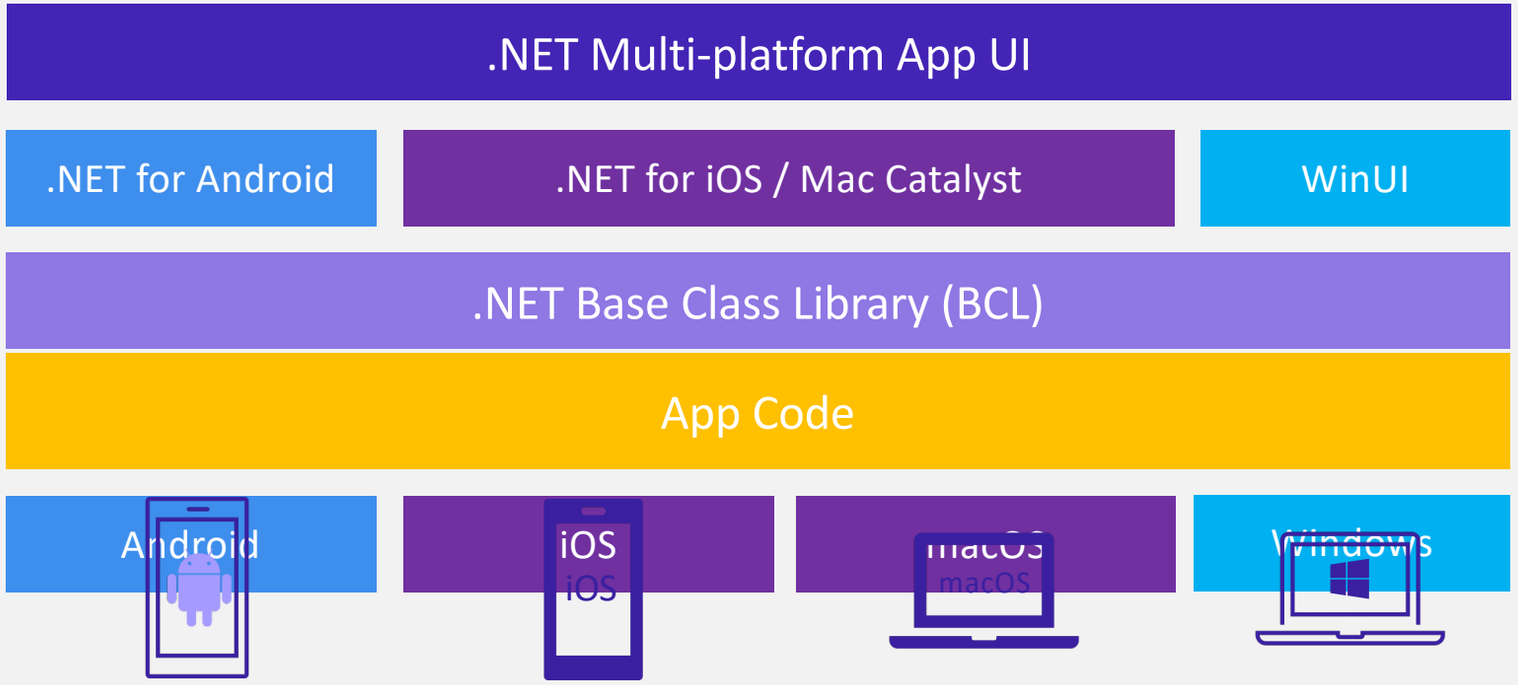
New to .NET are students

Download survey

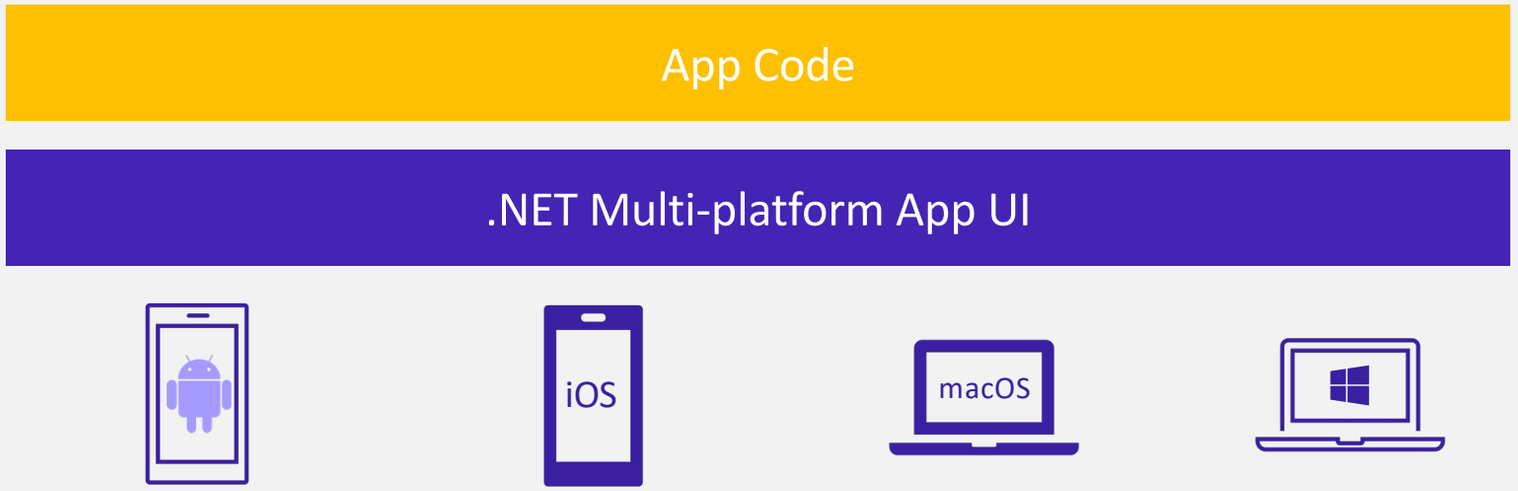
Sources: insights.stackoverflow.com/survey/2019#most-loved-dreaded-and-wanted; insights.stackoverflow.com/survey/2020#most-loved-dreaded-and-wanted; github.com/cncf/velocity/blob/master/reports/top_projects_by_activity.txt; octoverse.github.com/; www.techempower.com/benchmarks/#section=data-r20&hw=ph&test=plaintext

We are building .NET MAUI to be the most productive way to develop native apps that perform great on any device that runs Android, iOS, macOS, or Windows from a single codebase.

How does .NET MAUI work?



How does .NET MAUI work?



In the Box

- Android, iOS, macOS, Windows
- 40+ UI controls
- Modern navigation
- Menus
- Adaptive layouts
- Right-to-Left
- Semantic Accessibility
- Essential Platforms Services
- Asset Management
- Multi-targeting
- App Themes
- Multi-window Support
- App Builder

- Blazor integration
- CSS
- Shapes, Brushes, Gradients
- XAML features
  - Behaviors
  - Compiled bindings
  - Multi-bindings
  - Relative bindings
  - Styles
  - Value converters
  - Visual States
- Platform views
- Platform embedding
- Gestures
- Keyboard support
- Dual-screen

Component Vendors











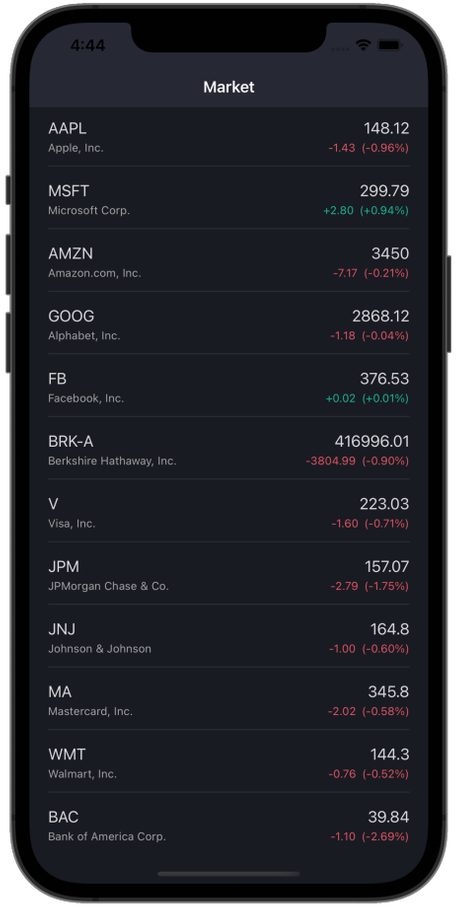


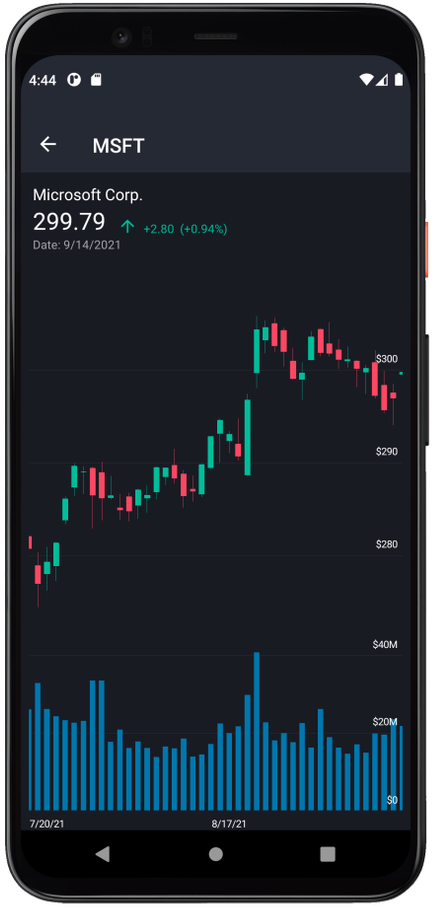




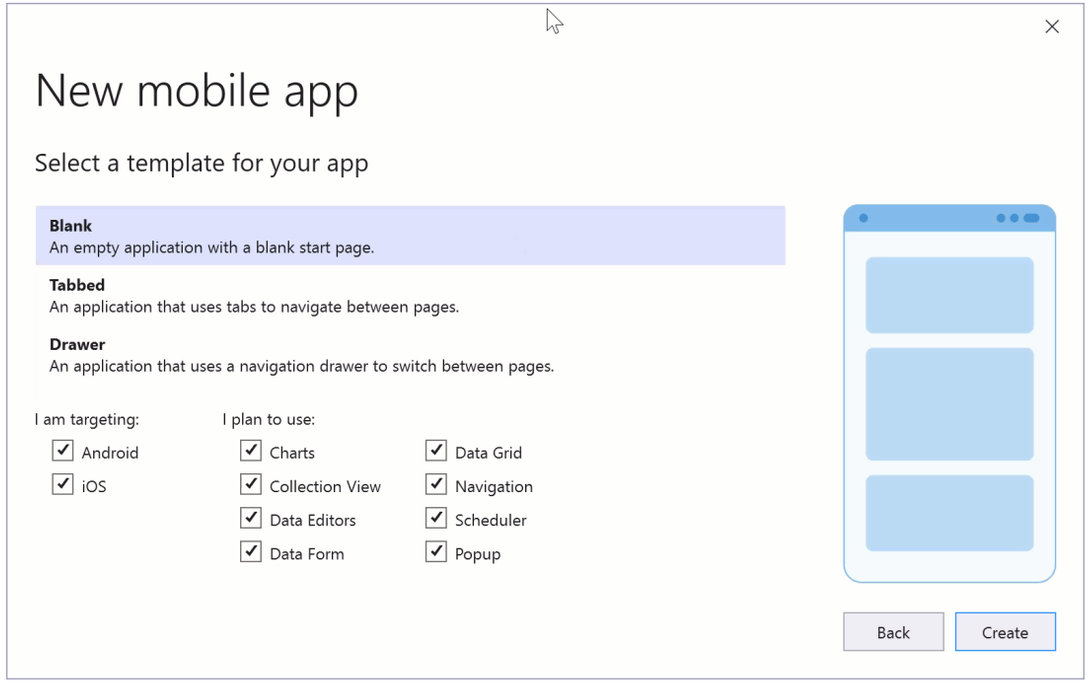


DevExpress





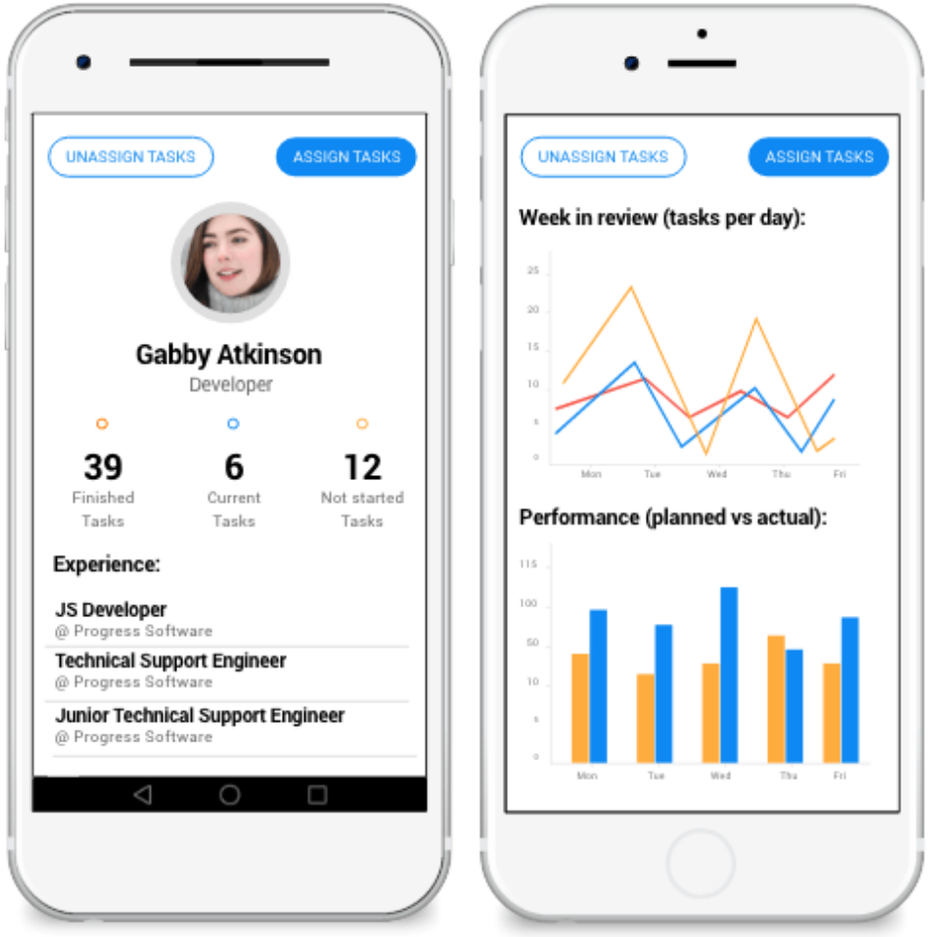




- CollectionView
- ChartView
- PieChartView
- DrawerView
- DrawerPage
- TabView
- TabPage
- DataGridView
- TextEdit
- PasswordEdit
- MultilineEdit
- ComboBoxEdit
- AutoCompleteEdit
- NumericEdit
- DateEdit / TimeEdit
- CheckEdit
- SimpleButton

https://community.devexpress.com/blogs/mobile/archive/2021/09/30/net-maui-free-early-access-preview-of-data-editors-and-more-for-mobile-development-v21-2.aspx

Progress Telerik

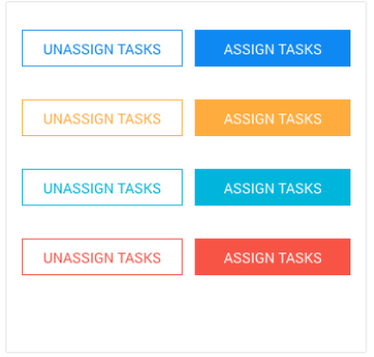


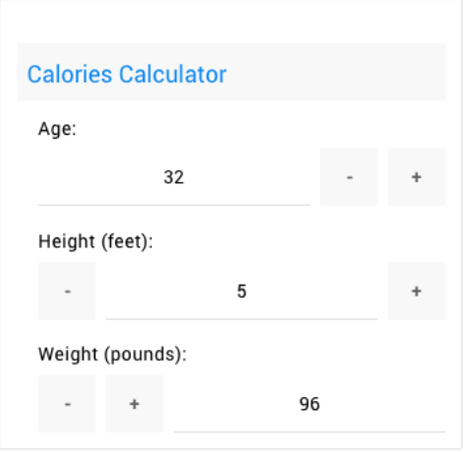




ID	Title	Assigned To
8207	UI Tests Builds timeout	Kian Parker
8211	Setup build machines	Davis Klein
8245	Add Barcode	Taylor Reed



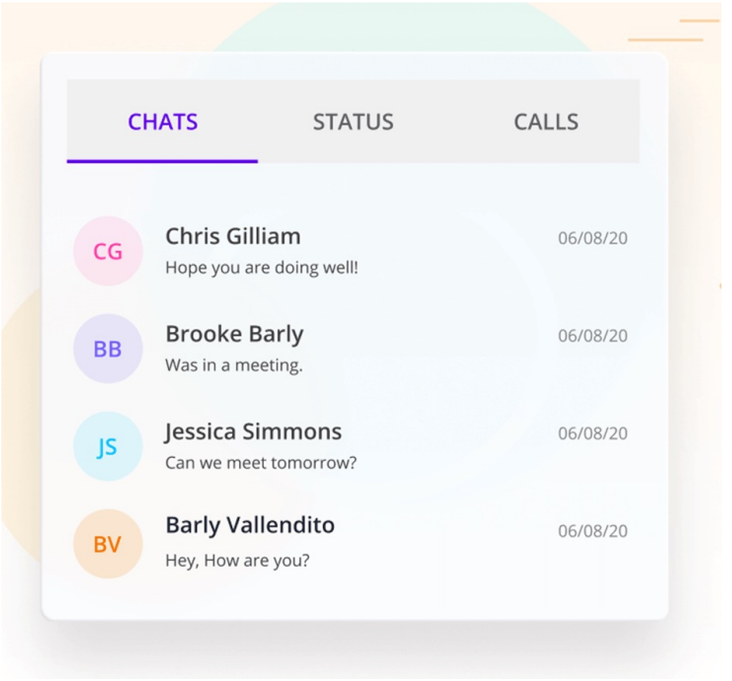




https://www.telerik.com/maui-ui

Syncfusion

[Introducing the First Set of Syncfusion .NET MAUI Controls | Syncfusion Blogs]





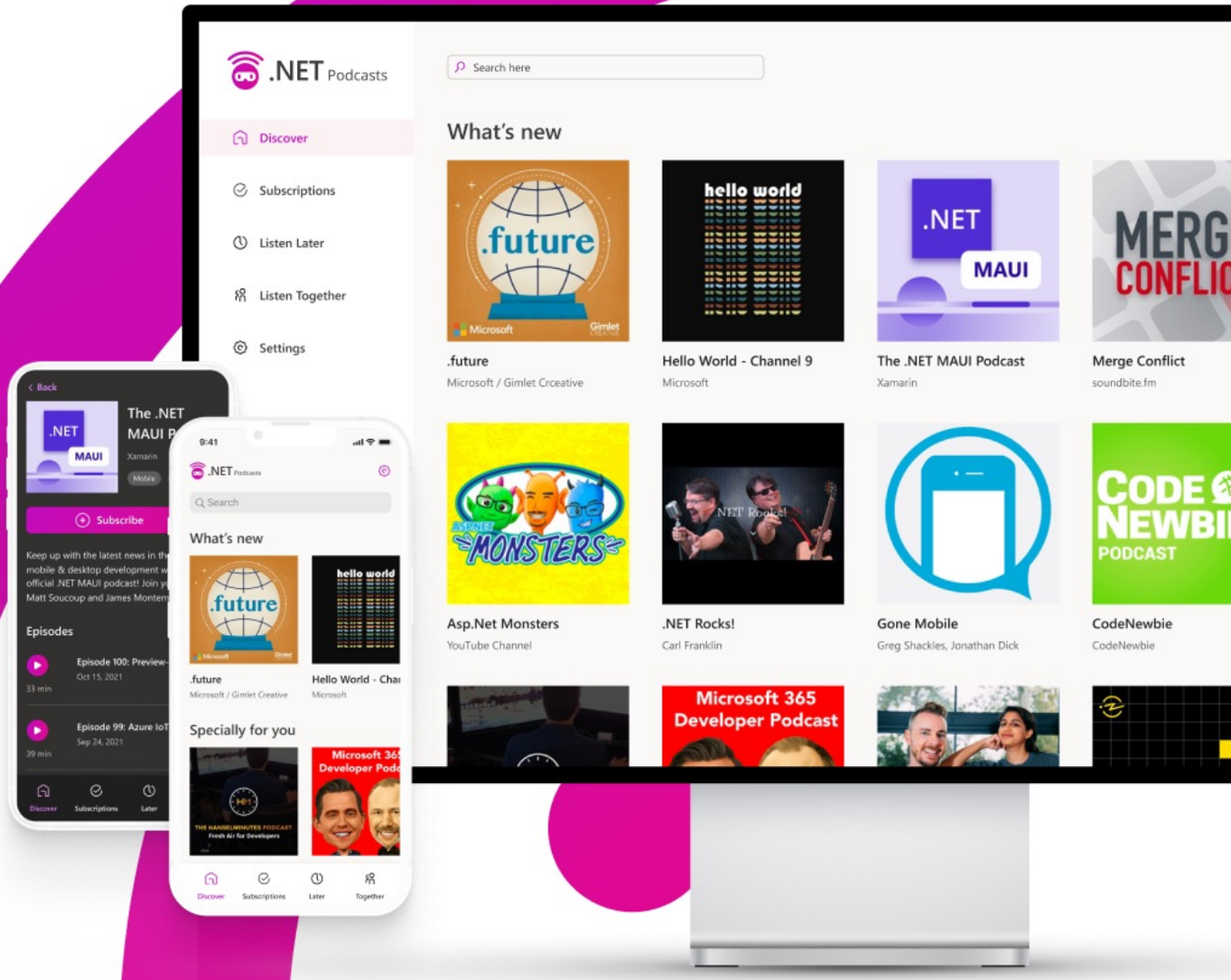


Are these controls migrated from Xamarin.Forms?
No. These controls were developed from scratch with the .NET MAUI graphics library and framework layouts, themselves with improved APIs and performance.

https://www.syncfusion.com/maui-controls

.NET Podcasts





https://github.com/microsoft/dotnet-podcasts



https://github.com/davidortinau/WeatherTwentyOne

Upcoming Milestones

- May GA for .NET MAUI
  - Best for new applications
  - Libraries should update and ship .NET 6 compatible versions
- May .NET Upgrade Assistant
  - Begin investigating Xamarin to .NET 6 migration
- June start previewing in .NET 7
- November GA of .NET MAUI vNext

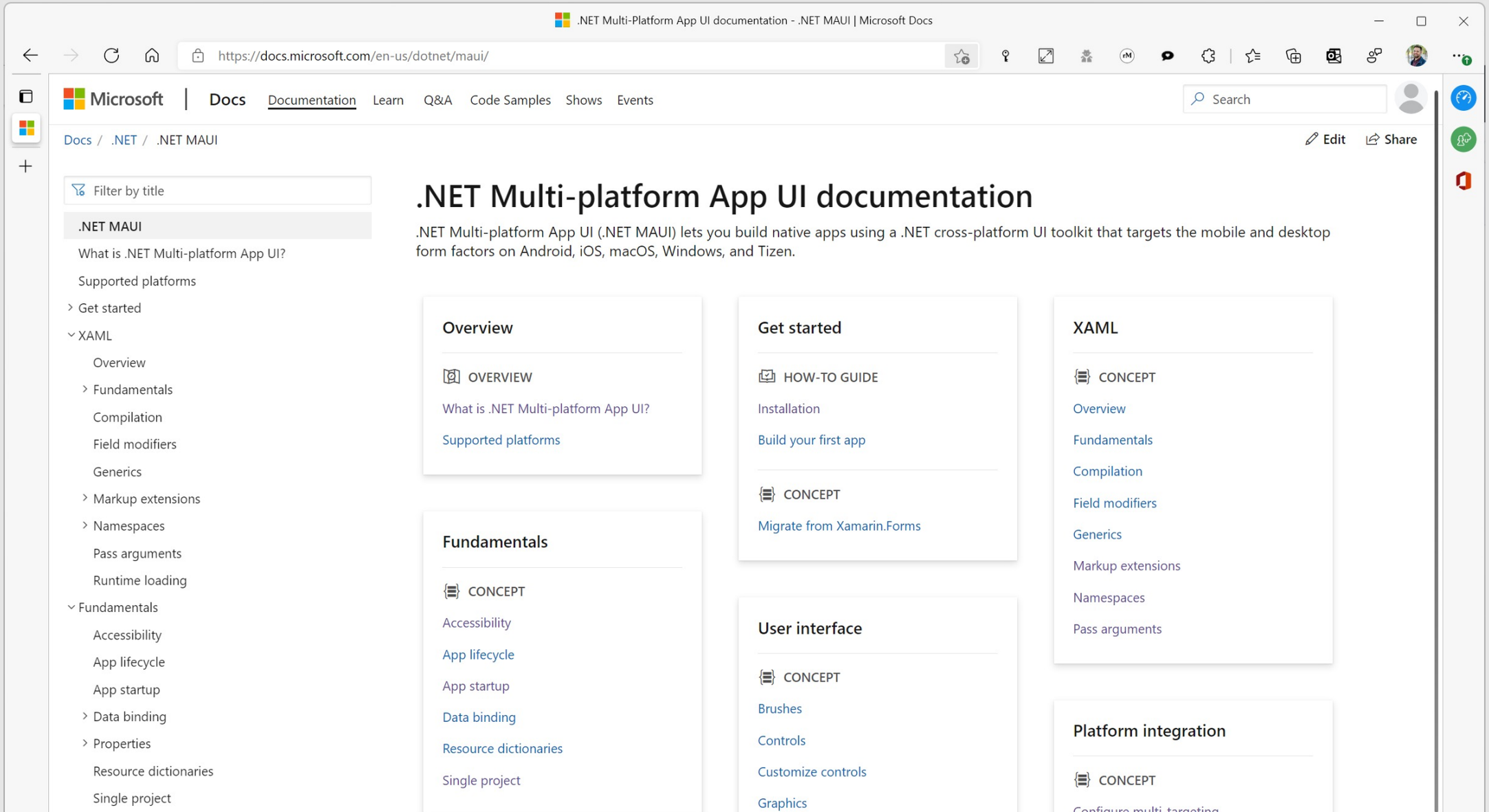
.NET MAUI Roadmap

Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov
• P14	• RC	• GA	• .NET 7 Preview • .NET MAUI SR					• .NET 7 GA

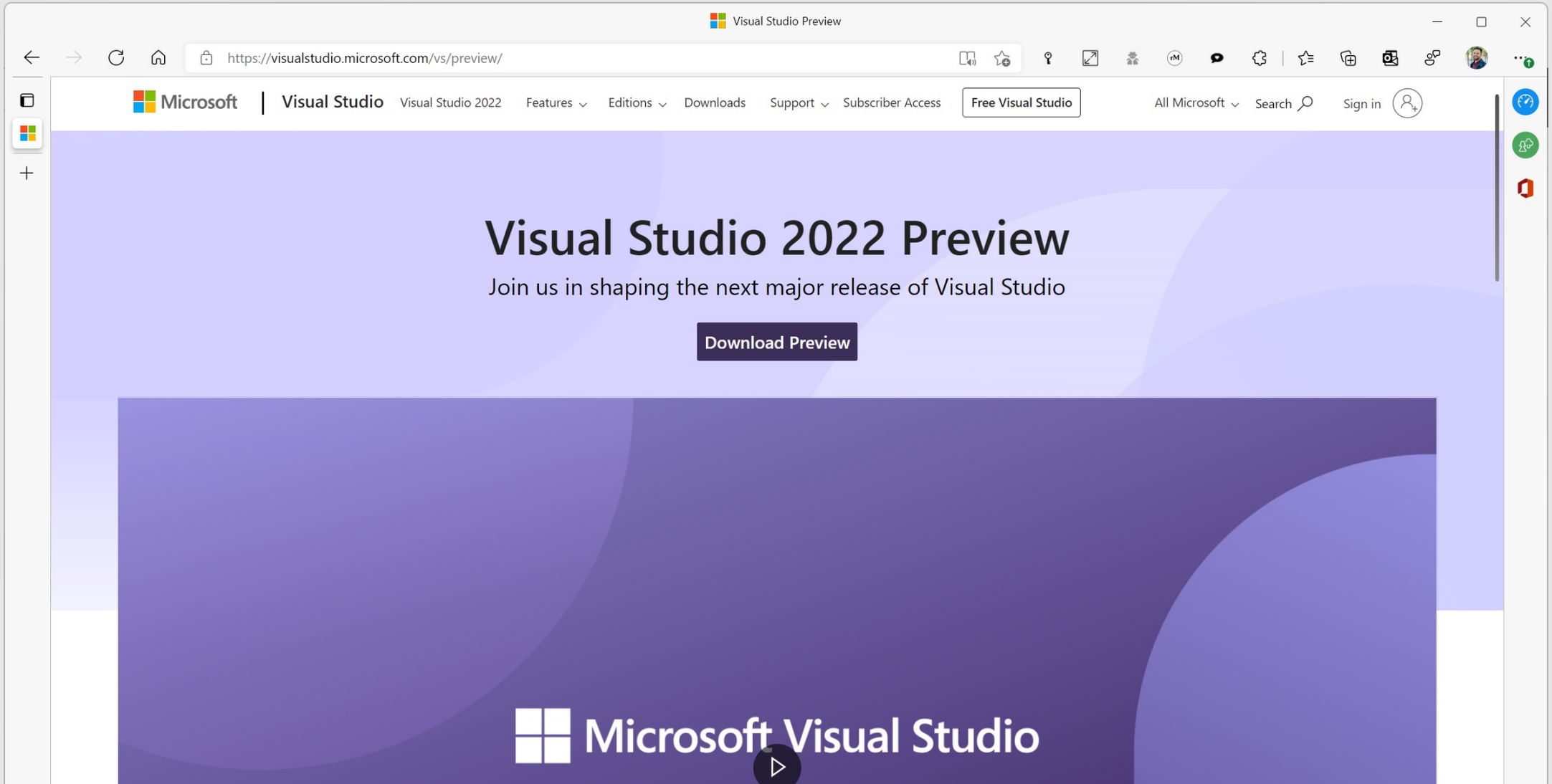
Migration

- Xamarin applications can all upgrade to .NET 6/7 without rewrite
- .NET Upgrade Assistant will
  - Change the *.csproj to SDK Style
  - Update Xamarin.Forms namespaces
  - Evaluate NuGet dependencies for compatibility with .NET 6
- Xamarin Android and iOS should at least be recompiled
  - .NET 6 aligns types that impact iOS apps – recompile
  - Android .NET Standard dependencies may work as-is
- Xamarin.Forms dependencies should all be replaced with .NET 6 compatible options
- After upgrade tasks:

Documentation is Growing



Get started today!



Microsoft

Any questions?

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